

Elimination of Fluoroscopy Use in a Pediatric Electrophysiology Laboratory Utilizing Three-Dimensional Mapping

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Objectives: The aim of this study was to quantify fluoroscopy use in catheter ablation procedures using a three-dimensional mapping system as the primary source of catheter guidance.

Background: Three-dimensional mapping allows continuous visualization of the location of mapping and ablation catheter electrodes. It has been shown to decrease fluoroscopy times. However, the extent to which it can decrease fluoroscopy time has not been completely defined.

Methods: Thirty patients (mean age 12.9 years; range 4–27 years) with reentrant supraventricular tachycardia underwent catheter ablation using standard protocols. Mapping was performed using the EnSite system (St. Jude Medical, St Paul, MN, USA) in the NavX mode (NavX). Eighteen patients had AVNRT, 12 had AVRT. Fluoroscopy times were compared to an age-matched and rhythm-matched control population.

Results: Procedural success was achieved in 30 of 30 patients (100%). Mean procedure time was 3.27 hours (range 1.83–5.8 hours). Mean fluoroscopy time was 1.05 minutes (range 0–14.8 minutes). Twenty-four of 30 (80%) received no fluoroscopy. Mean fluoroscopy time for the control group was 21.37 minutes (range 5.13–77.13 minutes). Thus fluoroscopy time was 95% less in the study group compared to control (1.05 ± 2.96 vs 21.37 ± 18.35 minutes, $P < 0.001$).

Conclusions: NavX was used to effectively guide catheters during ablation procedures resulting in a significant decrease in fluoroscopy use. In 80% of the procedures, no fluoroscopy was used. Further advances in the technology may permit additional decreases in x-ray exposure for the 20% of patients who required it. (PACE 2007; 30:510–518)

ablation, pediatrics, mapping, electrophysiology, clinical

Introduction

Radiation exposure related to diagnostic or therapeutic medical procedures carries known associated risks.^{1–3} These include dermatitis, cataracts, birth defects, and malignancies. These risks accrue to the patient as well as to the staff performing the procedure and are dose-dependent and cumulative over time. An expert consensus document on “Radiation Safety in the Practice of Cardiology,”⁴ published by the American College of Cardiology, recommends that all catheterization laboratory staff adopt the principle of ALARA (radiation doses “as low as reasonably achievable”). Multiple authors have published methods to minimize x-ray exposure during catheterizations.^{5–8} This is of particular relevance to the pediatric population due to the longer life expectancy of the patient. Many young patients with congenital defects will require multiple procedures for diagnostic or

therapeutic purposes. This can result in significant lifelong cumulative radiation exposure. Therefore, any steps taken to minimize the radiation dose will benefit both the patient and the staff.

Three-dimensional electrophysiologic mapping systems have been in use since the late 1990s. One of the benefits of this technology is to decrease x-ray exposure by giving the electrophysiologist a nonfluoroscopic tool to visualize the location of the catheter electrodes. The effectiveness of these systems for this purpose has been well documented.^{9–11} In our laboratory, we have been using the Ensite system (St. Jude Medical, St Paul, MN, USA) for catheter guidance since early 2003. For all uncomplicated supraventricular tachycardias (SVTs) it is used in the NavX mode (NavX). After more than 2 years experience with NavX, it has become our primary imaging modality for catheter ablation, with fluoroscopy functioning as an adjunct. This report presents successful reduction or elimination of fluoroscopy use in a large group of patients with reentrant SVT.

Methods

Patient Selection

The ablation registry at Akron Children's Hospital was searched for all patients who had undergone catheter ablation for reentrant SVT during

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the period from December 2005 through May 2006. During that period, 30 patients underwent the procedure. The patient charts were reviewed for clinical data including age, height, weight, tachycardia mechanism, procedure duration, fluoroscopy time, type of ablation performed (radiofrequency or cryoablation), total number of lesions, total lesion time, success, and complications. Although the majority of patients were in a pediatric age range, age was not a criterion for entry into this review. The only criterion for inclusion was the presence of reentrant SVT. All patients had an echocardiogram to document normal cardiac and vascular anatomy prior to procedure. Written informed consent was obtained for each procedure. Antiarrhythmic medications were discontinued for at least 5 half-lives prior to procedure. For a control group, the same hospital registry was searched for patients undergoing the procedure just prior to our first use of NavX in 2003. Patients were matched for age and tachycardia mechanism. To get a matched control group required including patients from the database between 2001 and 2003. During this time period we were not yet performing cryoablation.

Procedure

All procedures were performed under general anesthesia. A transesophageal pacing catheter was placed and positioned to give the largest obtainable atrial signal. This catheter was used as the stable reference catheter for the NavX system. Additional mapping catheters and the ablation catheter were then placed from a femoral venous approach. Electrophysiology study was performed using standard protocols. If the tachycardia mechanism could not be defined at baseline the patient received Isuprel prior to ablation. In all cases the patient was restudied both at rest and on Isuprel postablation. For patients with left-sided pathways requiring transseptal puncture, IV heparin was given at a dose of 100 units/kg up to a maximum of 10,000 units. Additional heparin boluses were given to maintain activated clotting times greater than 250 seconds.

Three-Dimensional Geometry

For placement of the venous catheters, the NavX system was focused at the femoral vein. The catheter could be seen as it exited the femoral sheath and entered the vein. The catheter was guided up the IVC. Entrance into a renal or hepatic vein could be directly visualized and, under NavX guidance, it could be withdrawn and redirected up the IVC. Intracardiac location of the catheter was confirmed by the presence of atrial electrograms. The point at which the atrial electrograms were first seen was marked on NavX as the IVC-

right atrial junction. The catheter was then advanced through the right atrium and into the SVC. The point at which atrial electrograms disappeared was marked on NavX as the SVC-right atrial junction. With these two reference points located, a right atrial geometry was created. The multipole catheter was maneuvered around the right atrium recording various points. Creation of an adequate, 3D image of the right atrium could be completed in several minutes. Next, the tricuspid valve was marked on the geometry. The tricuspid annulus was located by electrograms having equal atrial and ventricular signals. The catheter was then positioned to record a His electrogram. Once the His electrogram was identified, the catheter was shadowed on NavX to give a continuous marker of the His bundle. Finally, the catheter was positioned in the coronary sinus and a shadow of coronary sinus location was marked. The resulting 3D image is shown in Figs. 1 and 2. The image can be viewed from any angle and at any magnification. With the cavoatrial junctions, tricuspid valve, His bundle, and coronary sinus all marked, all catheters could be freely maneuvered and visualized within the right atrium. From this point on, the electrophysiology (EP) study and ablation were performed using standard protocols and procedures.

At this institution, cryoablation is the preferred energy source for all reentrant SVTs. In the current study, all but two patients had cryoablation as the energy source of first choice. The majority of patients had cryoablation performed using a 4-mm tip Freezor catheter (Cryocath Inc., Kirkland, Quebec, Canada). In these patients, a cryomap was defined as a preset temperature of -30°C to -50°C for 1 minute or less. A cryoablation was defined as a preset temperature of -60°C to -80°C for any length of time. In the remaining patients undergoing cryoablation, a 6-mm tip catheter was used. In these patients, a cryomap was defined as a lesion lasting less than or equal to 1 minute. A cryoablation was defined as any lesion lasting greater than 1 minute. In these patients, achieved catheter tip temperatures were -70°C to -80°C . If a cryomap site became a cryoablation, this site was considered a cryoablation and not counted as a cryomap. For the control group, radiofrequency energy was used in all patients. This occurred because we began using NavX in 2003, shortly before we began using cryoablation. Therefore, all patients in whom NavX was not used could not have received cryoablation.

Procedural Success

Procedural success was defined according to tachycardia mechanism. In patients with atrioventricular (AV) nodal reentrant tachycardia (AVNRT), success was defined as no inducible

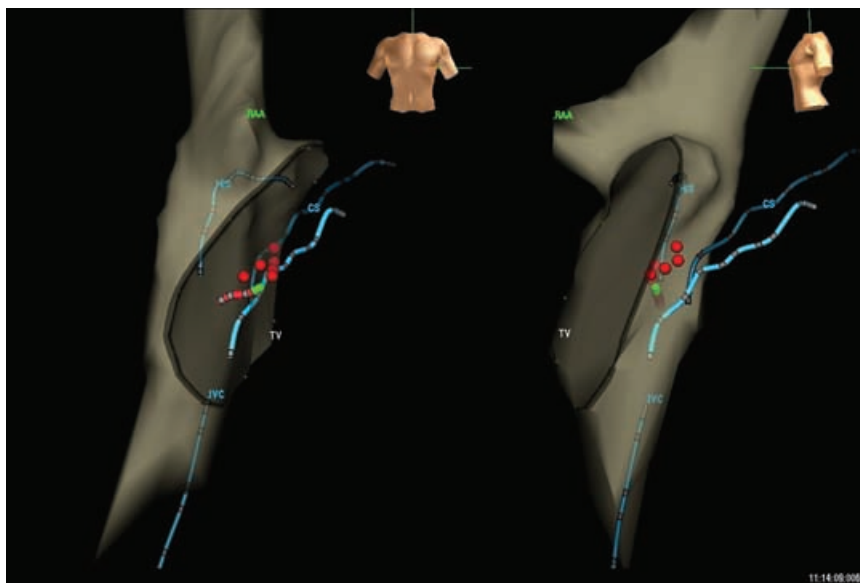


Figure 1. Three-dimensional geometry as it appears during ablation. Anteroposterior (left) and lateral view (right). CS = catheter shadowing location of coronary sinus. His = catheter shadowing the location of the His bundle, with the distal tip marking the area of His bundle recording. IVC = inferior vena cava; RAA = right atrial appendage; TV = tricuspid valve. The blue catheter is a decapolar mapping catheter, which at the time of this image is located within the coronary sinus. The red and green catheter is the ablation catheter. Red spheres mark the location of cryolesions in this patient with AVNRT.

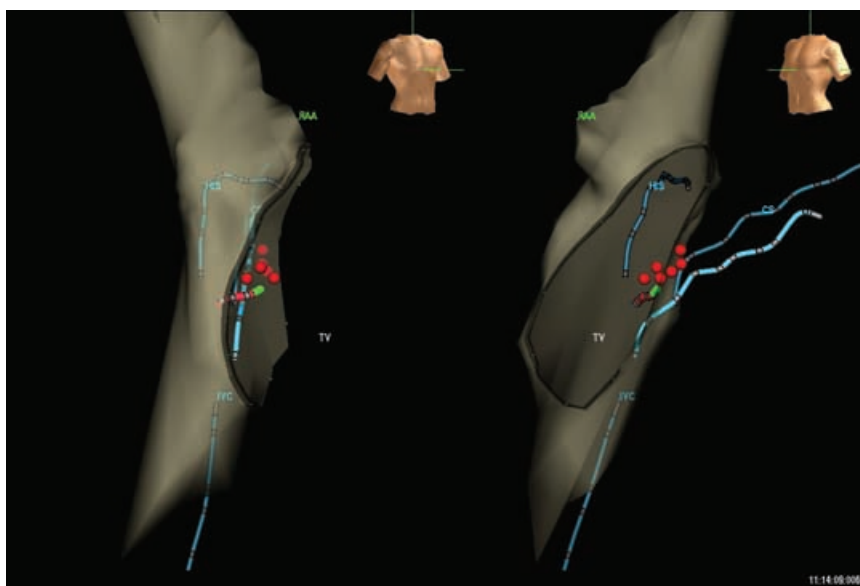


Figure 2. Three-dimensional geometry as it appears during ablation. Right anterior oblique (left) and left anterior oblique view (right). CS = catheter shadowing location of coronary sinus. His = catheter shadowing the location of the His bundle, with the distal tip marking the area of His bundle recording. IVC = inferior vena cava; RAA = right atrial appendage; TV = tricuspid valve. The blue catheter is a decapolar mapping catheter, which at the time of this image is located within the coronary sinus. The red and green catheter is the ablation catheter. Red spheres mark the location of cryolesions in this patient with AVNRT.

SVT either at rest or on Isuprel for greater than 30 minutes postablation. Isolated echo beats were acceptable, but not more than single echo beats. For accessory pathway-mediated tachycardia, success was defined as no inducible SVT, either at rest or on Isuprel, for at least 30 minutes beyond the last lesion. In addition, we documented transient AV block with adenosine. In patients with concealed pathways, transient block was documented in a retrograde fashion. In patients with manifest preexcitation, transient AV block was documented both antegrade and retrograde. If a patient had retrograde conduction postablation, then pacing protocols were performed to document that the location of retrograde conduction was within the atrial septum and that it had decremental retrograde properties consistent with conduction through the AV node. Procedure time was defined as the time interval from initial femoral venous sheath placement until femoral venous sheath removal.

Statistical Analysis

Values are shown as mean $\pm$ standard deviation unless otherwise specified. Comparisons were made with an independent samples T test or a Mann-Whitney two sample test, as appropriate. A P value of less than or equal to 0.05 was considered statistically significant.

Results

Patients

Thirty patients (16 females, 14 males) are included in this retrospective review (Table I). Mean age was 12.9 ± 4.9 years; height 157 ± 23 cm; and weight 54 ± 21 kg. Although the age range was 4–27 years, only one patient was over 18 years. Eighteen patients had AVNRT and 12 patients had atrioventricular reentrant tachycardia (AVRT). Among patients with AVRT, accessory pathway locations were as follows: two right anteroseptal, three right posteroseptal, one right lateral, three left posteroseptal, two left posterolateral, and one left anterolateral. Nine of the accessory pathways showed manifest preexcitation and three were concealed. All patients had structurally normal hearts and one patient with wolffparkinson-white (WPW) had a dilated cardiomyopathy. One of the patients had undergone two previous catheter ablation attempts. In the control group there were 19 females and 11 males. Mean age was 12.3 ± 4.8 years; height 154 ± 19 cm; and weight 51 ± 18 kg. Eighteen patients had AVNRT and 12 had AVRT.

Success

Procedural success was achieved in 30 out of 30 of the study patients. For the 18 patients with

AVNRT, slow pathway conduction was completely eliminated in 16 patients and modified in the remaining 2 patients. Sixteen patients had no echo beats and no evidence of dual AV nodal physiology postablation. Of the 28 patients undergoing cryoablation, 27 (96%) had successful elimination of their tachycardia mechanism with cryoablation alone. As previously stated, one patient with a right free-wall manifest accessory pathway failed cryoablation and had radiofrequency ablation performed during the same setting. For those undergoing cryoablation, there was a mean number of cryomaps of 8.3 with a range of 0–30. The mean number of cryoablation lesions was 5.9 with a range of 2–13. The total cryoablation time averaged 21 minutes with a range of 8–52 minutes. Two patients with left-sided accessory pathways had radiofrequency ablation as the sole energy source. In both of these patients it was felt that, due to the patient's size and pathway location, the larger stiffer cryocath would be difficult to maneuver to the target site, and therefore, radiofrequency (RF) was chosen as the primary tool. For those three patients receiving radiofrequency ablation, the mean number of RF lesions was 24 with a range of 6–35. The total RF time was 11 minutes, with a range of 5.5–16 minutes.

In the control group, acute success was achieved in 29 out of 30 patients.

Fluoroscopy Time

Mean fluoroscopy time for the control group was 21.4 ± 18.3 minutes. Mean fluoroscopy time for the study group was 1.05 ± 2.9 minutes (Table II). Thus, the group in which NavX was used received 95% less fluoroscopy than the control group (21.4 vs 1.05 , $P < 0.001$). Only 6 of the NavX patients received any fluoroscopy; the remaining 24 patients (80%) received none. Fluoroscopy times for the 6 patients who received it ranged from 1.2 to 14.8 minutes, with a mean of 5.3 ± 4.9 minutes. When compared to the control group, this also was significantly less (21.4 vs 5.3 , $P = 0.001$). None of the 18 patients with AVNRT required fluoroscopy. Six of the 12 patients with AVRT did require fluoroscopy. Of these, 5 had left-sided accessory pathways which necessitated the use of fluoroscopy to perform a transseptal puncture. Only one patient required fluoroscopy who did not also require a transseptal puncture. This patient had a right free-wall accessory pathway.

Procedure Time

Mean procedure time was 3.27 hours with a range of 1.83–5.73 hours. Mean procedure time for the control group was 3.17 hours with a range of 1.63–5.13 hours. Thus, there was no significant

Table I.

Demographics and Procedural Data of the NavX Group

Patient	Age (years)	Height (cm)	Weight (kg)	SVT Mechanism	Pathway Location	Number of Cryolesions	Total Cryoablation time (minutes)	Number of RF Lesions	Total RF Time (minutes)	Procedure Time (hours)	Fluoroscopy Time (minutes)
1	16	178	61	AVNRT		2	8	0	0	2.45	0
2	5	115	22	AVNRT		2	8	0	0	2.15	0
3	4	110	17	AVNRT		8	16.85	0	0	4.55	0
4	14	180	57	AVNRT		3	12	0	0	2.87	0
5	14	170	60	AVNRT		3	12	0	0	1.83	0
6	12	148	43	WPW	Right anteroseptal	5	12.87	0	0	2.87	0
7	6	133	51	AVNRT	Right lateral	4	12.12	0	0	2.33	0
8	11	150	33	WPW	Left posteroseptal,	12	5.93	35	16	5.18	4.98
9	15	173	67	WPW	Right posteroseptal	0	0	30	13	5.8	14.82
10	15	161	60	AVNRT		6	24	0	0	2.52	0
11	13	162	89	WPW	Right anteroseptal	2	4.47	0	0	2.4	0
12	10	151	48	WPW	Right posteroseptal	5	20	0	0	1.93	0
13	15	179	79	AVNRT		4	16	0	0	2.3	0
14	13	160	51	AVNRT		12	33.1	0	0	5.73	0
15	13	170	70	Concealed pathway	Right posteroseptal	5	20	0	0	2.33	0
16	6	118	21	AVNRT		3	12	0	0	1.92	0
17	16	170	74	AVNRT		4	12.58	0	0	2.72	0
18	10	126	27	Concealed pathway	Left posterolateral	6	24	0	0	4.5	2.55
19	18	169	64	AVNRT		4	13.22	0	0	2.08	0
20	16	192	82	WPW	Left posterolateral	9	36	0	0	3.9	1.2
21	18	173	63	AVNRT		6	24	0	0	2.45	0
22	8	135	25	WPW	Left anterolat	11	46.02	0	0	4.2	2.55
23	12	173	64	AVNRT		3	12	0	0	2.18	0
24	10	139	27	AVNRT		9	14.26	0	0	4.18	0
25	17	177	96	Concealed pathway	Left posteroseptal	9	36	0	0	3.95	5.38
26	6	122	23	WPW	Left posteroseptal	0	0	6	5.52	4.43	0
27	18	175	64	AVNRT		11	36.85	0	0	4.53	0
28	15	161	56	WPW	Right posteroseptal	5	20	0	0	3.46	0
29	15	175	60	AVNRT		13	52	0	0	2.75	0
30	27	171	68	AVNRT		12	43.63	0	0	3.6	0

AVNRT = atrioventricular (AV) nodal reentrant tachycardia; WPW = Wolff-Parkinson-White.

Table II.
Results of Study Group versus Control

	NavX Group, Mean, (SD)	Control Group, Mean (SD)	P Value
N	30	30	NS
Age (years)	12.9 (4.9)	12.3 (3.8)	NS
Weight (kg)	54.1 (21.4)	52 (18.4)	NS
Procedure time (hours)	3.27 (1.18)	3.17 (1.13)	NS
Fluoroscopy time (minutes)	1.05 (2.96)	21.37 (18.35)	<0.001
Acute success	30/30	29/30	NS
Recurrence	4/30	3/29	NS

difference in procedure times between the two groups ($P = 0.754$).

Complications

Some form of transient AV block was seen during 10 applications of cryotherapy. Nine patients had transient second- or third-degree AV block and one patient had transient right bundle branch block (RBBB). Both patients with antero-septal accessory pathways had transient complete AV block. Seven patients with AVNRT had transient AV Block. The patient with RBBB had AVNRT as his tachycardia mechanism. In all patients the AV block resolved quickly after discontinuation of the freeze. There was no long-term AV block and no other complications.

Recurrence

During follow-up of 3 months (range 1–6 months), 4 of 30 patients (13%) had a recurrence. Two of these patients had AVNRT, and 2 patients had AVRT. In the control group, 3 of 29 patients (10.3%) had a recurrence. In both groups, freedom from tachycardia was 87% at 2 months postprocedure (26 out of 30 patients). In the control group there were no further recurrences at 1-year follow-up.

Discussion

In the national pediatric ablation registry the average fluoroscopy time for ablation of SVT is 40 minutes.¹² Fluoroscopy times any greater than average are associated with a small but measurable increase in the incidence of malignancy.¹³ This risk is lifelong and therefore greater in younger patients. The ALARA principle was published to remind all physicians and staff that radiation exposure should be kept to a minimum. Methods to achieve this goal include using lower image frame rates, lower resolution imaging, and stepping on the fluoroscopy pedal only as long as necessary.

In the present study NavX was used as the sole imaging modality in the majority of patients. It must be noted, however, that this was after more than 2 years experience using and learning the system in conjunction with fluoroscopy. NavX presents data in a manner similar but not identical to fluoroscopy. There is a learning curve necessary for the user to become accustomed to this new way of viewing catheters, as well as for the more technical adjustments necessary with the equipment. As our own confidence with the system increased, a deliberate approach developed to maximize our utilization of NavX and minimize fluoroscopy. The present study is a result of that deliberate bias. In 80% of the procedures NavX eliminated fluoroscopy use. In the remaining patients it decreased fluoroscopy use by 75%. This degree of radiation dose reduction has the potential to significantly decrease long-term risks to both patient and staff.

Catheter ablation without the use of fluoroscopy was first reported by Drago et al. in 2002.¹⁴ In that study the authors used CARTO (Cordis Webster, Marlton, NJ, USA) as their navigation tool and radiofrequency as their energy source. They limited their patient selection to right-sided free-wall accessory pathways. More recently, Grubb et al. have reported the use of NavX to minimize or eliminate fluoroscopy in adult patients.¹⁵ NavX has several advantages for a nonfluoroscopic approach. First, the system visualizes all catheters that are used during the procedure, not just the ablation catheter. Second, any EP or ablation catheter can be used and therefore either cryoablation or radiofrequency can be chosen for the procedure. Cryoablation offers a safer approach to tachycardia substrates on the atrial septum in children.^{16–18} Because of this, NavX allows for ablation in these areas without the use of fluoroscopy. Since catheters can be positioned in the His bundle and coronary sinus and visualized continuously on NavX, Koch's triangle is easily defined and precise location of lesion delivery can be tightly controlled. It

is our practice that all tachycardia substrates near the AV node be approached first with cryoablation. This is particularly important in procedures done without fluoroscopy since it confers a greater degree of protection against permanent heart block.

It is of particular interest that none of the patients with AVNRT in this study required fluoroscopy. In fact, of the 25 patients whose tachycardias were ablated from the right atrium, 24 (96%) received no fluoroscopy. One patient with a right atrial free-wall manifest accessory pathway required fluoroscopy. In this patient, initial attempts at cryoablation of the pathway failed and radiofrequency was then attempted. However, initial attempts at radiofrequency ablation also failed. We therefore switched to shaped sheaths to assist with catheter tip positioning against the pathway. Since the shaped sheaths cannot be seen on NavX, this necessitated the use of fluoroscopy. Sheaths with electrode tips would be very useful in these situations.

The most common reason for fluoroscopy use in this group of patients was the need to perform a transseptal puncture. This was the case in 5 of the 6 patients requiring fluoroscopy. An area of particular research interest is to evaluate methods of transseptal puncture that do not require x-ray exposure. Intracardiac echocardiography is a well-known adjunct to perform a transseptal puncture,¹⁹ and this may be the best approach for eliminating fluoroscopy entirely. Transesophageal echocardiography is easier to position without fluoroscopy and gives a greater range of views of the heart. It may, therefore, also be a reasonable alternative. Verma and Borganelli reported the use of NavX to visualize a Brockenbrough needle during transseptal puncture.²⁰ The addition of electrode tips to the sheath and dilator would greatly enhance the ease with which a transseptal puncture could be performed. Detailed mapping of the atrial septum may delineate the fossa ovalis, and echocardiography can confirm successful placement. If a reliable method for transseptal puncture can be developed then fluoroscopy during catheter ablation may become an uncommon to rare occurrence.

In addition to decreased radiation exposure, NavX guidance for catheter ablation offers several other potential benefits. The first is cath lab staff comfort. In our laboratory the staff no longer wears lead aprons during the procedure. In the few patients requiring transseptal puncture, a lead apron is donned for the short period during which the transseptal puncture is being performed. Thereafter the lead can be removed for the remainder of the case. Another potential benefit of NavX is portability. Because all of the equipment involved is relatively small and portable, it is now quite fea-

sible to do an ablation in the ICU rather than transporting a sick patient to the cath lab. A bedside ablation procedure could be easily performed with NavX, utilizing portable fluoroscopy if necessary. Finally, NavX guidance offers the possibility of doing ablations at times that would otherwise have been precluded. Specifically, an ablation could be performed on a female during pregnancy without significant risk of x-ray exposure to the fetus.

None of the patients in this study had significant congenital heart disease. Performing an ablation without fluoroscopy in a patient with complex heart disease would be a more difficult task to undertake. However, it would still be expected that NavX could decrease fluoroscopy times in those patients due to the ability to see the catheters in real-time 3D. Papagiannis et al. recently reported both a decrease in fluoroscopy time and a decrease in overall procedure time when NavX was used during catheter ablation in children.²¹ In the present study there was no significant difference in procedure time between the study group and controls. The most likely reason for this is that the study group received primarily cryoablation, while in the control group radiofrequency was used. A single radiofrequency lesion takes 1 minute, while a cryo lesion takes 4 minutes, thus adding time to the procedure. Additionally, in those cases in which NavX is used without fluoroscopy, more time is usually spent to generate an ideal geometry before proceeding to ablation.

There was a 13% recurrence rate by 2 months follow-up in the study group. Thus, at 2 months postprocedure, 87% were tachycardia free. This is not significantly different from the Prospective Assessment after Pediatric Cardiac Ablation database (PAPCA).²² In that group there was a 4% acute failure rate and 7% recurrence rate at 2 months. Thus, in the PAPCA group 89% were tachycardia free at 2 months postprocedure. Possible causes of recurrence include method of ablation (cryoablation vs radiofrequency) and accuracy of lesion location using NavX versus fluoroscopy. Long-term recurrence rates remain to be seen.

Potential areas of concern with a nonfluoroscopic approach include risk of perforation, risk of catheter knot, risk of heart block, and risk of geometry frame shift. The risk of perforation is present with any cardiac catheterization. However, it would not be anticipated that perforation would be more likely using NavX guidance versus fluoroscopy since, with NavX, there is continuous real-time biplane catheter imaging, compared to intermittent single-plane visualization using fluoroscopy. With cryoablation the risk of permanent AV block should be quite remote as long as care is taken to look for His signals on the ablation catheter prior to lesion delivery and AV

conduction is monitored closely during cryoablation. Transient AV block is a commonly reported finding^{16–18}; however, permanent heart block is rare. As previously stated, cryoablation is our first-line approach to all tachycardia substrates on the atrial septum. Catheter entanglement or catheter knot is a potential problem with NavX. Since NavX can only visualize the electrodes of the catheter and not the catheter shaft there is the potential for two catheters to become entangled. However, this would most likely be felt as a change in resistance to catheter manipulation, at which point the catheters could be visualized under fluoroscopy and, if there is entanglement, appropriate maneuvers to disentangle them could be performed. The final concern about NavX is geometry reference shift. This happens if the reference catheter moves inadvertently. In our study, we chose an esophageal catheter as the reference catheter. This eliminates the chance of dislodgement by contact with transvenous catheters. There remains a small chance of catheter movement related to peristaltic movement of the esophagus. However, we have not witnessed this in practice during our procedures. This is more likely to be an issue in a conscious patient as compared to an anesthetized patient. Other locations that have been used include the right ventricular apex and the coronary sinus. Although these catheter positions tend to be quite stable, they can be dislodged during manipulation of the ablation catheter. We, therefore, prefer an esophageal catheter. Ultimately, a surface electrode may be the best solution for this problem.

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The surface electrode can be marked in location so that even if the lead is dislodged it can be repositioned in the exact location.

Study Limitations

This study has a relatively small number of subjects reported. Because of this, complications which are uncommon or rare cannot be fully evaluated. A much larger sample size will be necessary to evaluate whether or not this approach will have any impact on the uncommon or rare complications. Also, follow-up times are short and long-term success remains to be determined. This study is not a randomized comparison of NavX to fluoroscopy. The authors acknowledge a deliberate bias toward the use of NavX and away from the use of fluoroscopy. Finally, any conclusions made regarding the differences between the NavX group and controls must be made with the awareness that the two groups also differed in their method of ablation.

Conclusion

NavX guidance can safely and effectively be used as the primary catheter visualization tool during catheter ablation. It decreases fluoroscopy use in all cases and in most right-sided tachycardia mechanisms it eliminates the need for fluoroscopy altogether. This will decrease the radiation risks to both the patient and the staff. Further research and technological advances are needed before it will replace fluoroscopy for left-sided tachycardias.

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